Engineering Report — Laurel Height Secondary School Electric Vehicle Club

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Abstract

After the cancelation of 24V race, the motor that LHSS EVC owns no longer fits our purpose of 12V race. A new drivetrain configuration , steering system, and vehicle chasis has to be created in order to maximize, speed, efficiency, and minimize weight for 805. Two Kraken X60 motors and a gearbox providing 12:1 ratio was purchased. By testing the vehicle on track, data shows positive results, generating a speed of up to 40 km/h, and having sufficient capacity to finish the race. Therefore, the new vehicle 805 is believed to succeed in our next race with the new design.

1 Introduction

1.1 Project Overview

The race vehicle, 803's performance in the Waterloo EV Challenge October race 2025 was not optimal. Although the vehicle leads the race in the first 15 minutes, the battery capacity deteriates rapidly, leading to the inability to finish the race.

1.2 Problem Statement

803 was equipped with one Saietta BR-119-R-68 motor, designed for 48V use. According to the technical specification provided by the manufacturer, the motor can create a torque of 12.1 Nm. Yet, due to the nature of 48V motor, it lacks the ability to provide efficient run in 12V useage. The motor consumes amperage over our expectation, and therefore leading to our early end in the race. Additionally, the motor weighs 11kg, increasing the weight of the car, therefore requires more energy to accelerate. Moreover, the vehicle is equipped with anti-ackermann steering system, which does not benefit our relatively low-speed race.

1.3 Design Objectives

After analysis from the preivous race, the following goals are created

1. To create new steering system that allows efficient turns

2. To increase the vehicle top speed to above 35 km/h
3. To increase the efficiency of drivetrain, and successfully finish the race (70 minutes)
4. To reduce weight of the vehicle

2 Background

2.1 Competition Overview

The competition occurs at the parking lot area around the East Campus of University of Waterloo. The race track is around 950m, with 4 major turns (Figure 1.1).

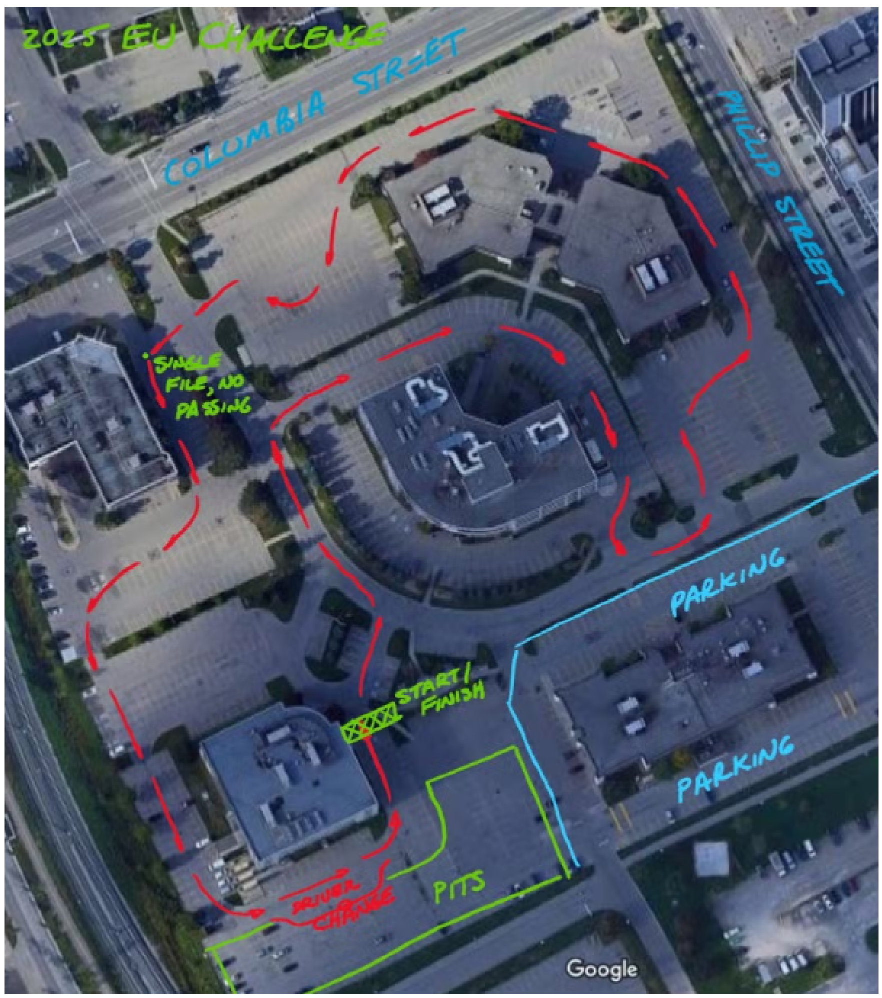

Figure 1: Course for May 2026 Race

2.2 Design Constraints

Rules for 2026 Races are as follows,

1. Distance between tires must be at least 0.90 m
2. Vehicle no wider than 1.2m
3. Maximum vehicle length is 3.5m
4. Driver must be in a seated, forward-facing position
5. Frame must has sufficient strucutal strength to protect the Driver
6. Two horizontal member spaced 6 to 10 inches apart to protect driver's legs
7. Steel / Aluminum
8. Roll bar must extend at least 50mm over the top of the helmet
9. The steering must allow for a turn within a radius of 5m
10. MTX-35 battery only
11. Battery must be secured
12. Driver must of complete manual control of the vehicle

3 Methodology

3.1 Frame

Materials

The perivous vehicle generations used steel as frames. However, such design would increase the weight of the vehicle. Therefore, the current design now accquires aluminum as the main material.

According to simulation, the weight is reduced form 64.4 kg to 22.5 kg. This significantly reduced the amount of energy needed to accelerate 805 to its top speed.

Design

805 shares the same frame with 804. The design seperate 805 into three sections: Leg, Body, Back.

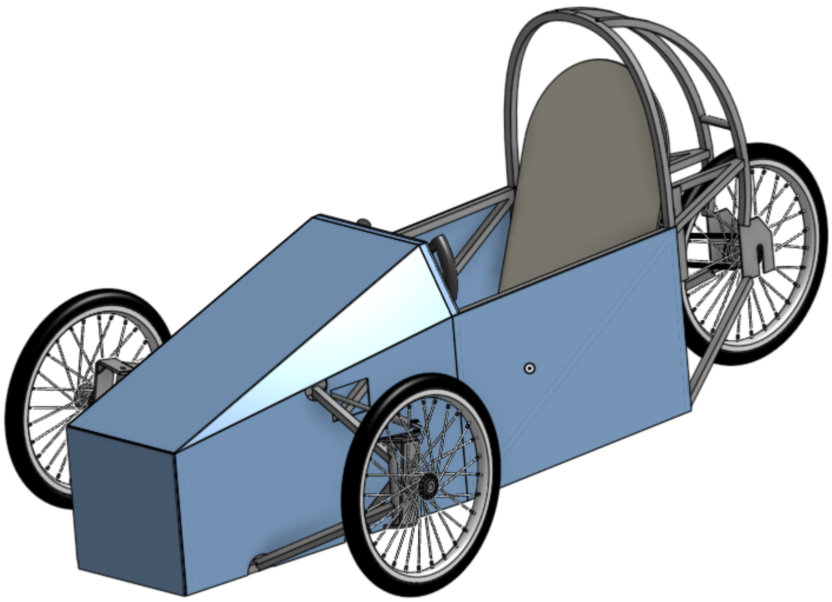

Figure 2: 805 Design.

The leg section is the front portion of 805. It provides room for driver's leg, steering systems, and a MTX-35 battery.

The battery is located at the most front of the car. This location allows for a more balanced weight distribusion across the car, counteracting the center of mass of the driver. The battery is held secure by the battery holder. This holder set contains two rods that bolds the battery at its position, with four-sided barriers at the bottom, preventing the battery from leaving its position.

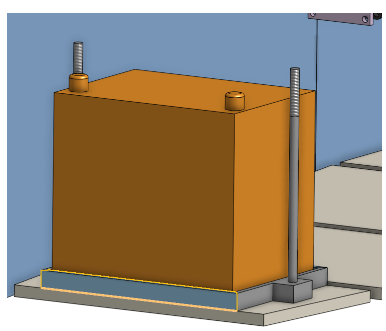

Figure 3: Battery Design.

A footrest is created in preventing the driver from knocking off the battery from the fixed location.

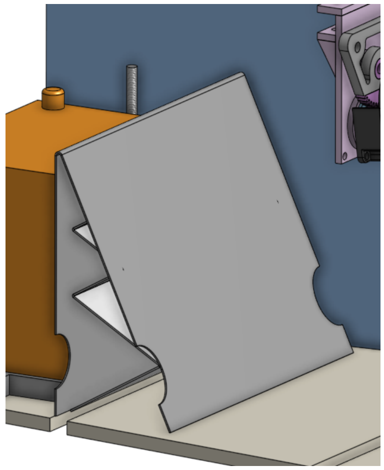

Figure 4: Footrest Design.

The steering gearbox is located at the middle-top crossbar, bolted in place. The specific details of this gearbox will be discussed in section 3.2.

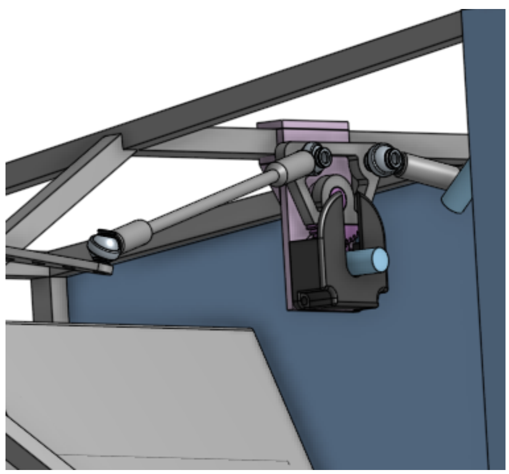

Figure 5: Steering Location Design.

The leg section is covered by three panels (top, left, and right). The design of these three panels is to streamline the vehicle, therefore reducing the drag coefficient.

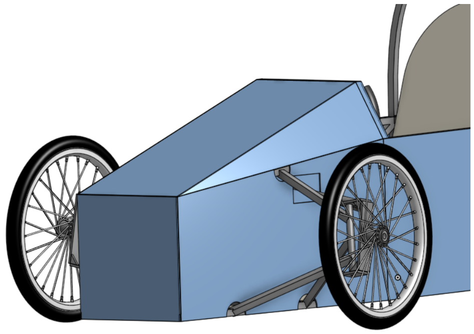

Figure 6: Front Panel Design.

The body section is where the driver sits and operates. This section contains the kill switch (in yellow), yoke, and driver seat.

The yoke is directly connected to three steering bar, with 2 universial joints in order to control the steering gearbox. The driver sits on a wood plate, which provides an incline that not only reduces the coefficient of drag, but to allow the driver to see a minimum distance of 2 meters in front of the car. The kill switch is on the right side of the driver, which provides easy access in case of an emergency.

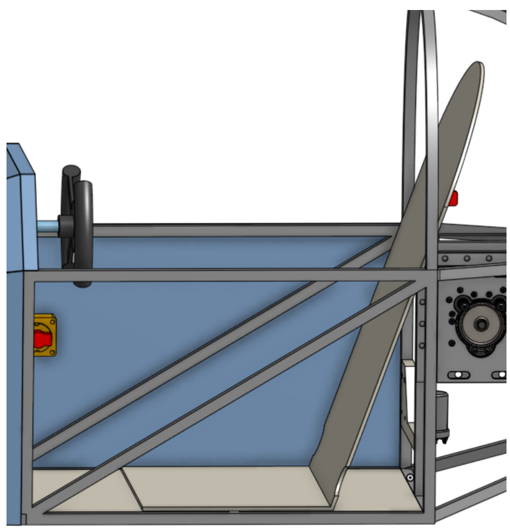

Figure 7: Body Section Design.

The back section of the car stores every compoenents of the drivetrain. Two Kraken X60s are mounted to a flipped gearbox, and the gearbox is attached to two mountaing plates. After a gear reduction of 9:1, the gearbox will then export the rotational energy to a sprocket that is connected to the rear wheel.

The two motors are connected via wires, to a CANBUS and Rasberry Pi control unit. They are stored and mounted on the panel located above the wheel.

A roll-cage is designed to start from the bottom of 805, and bends over the driver's helmet. A 50mm clearance is engaged in protection for the driver's safety. Two bent support beam is connected from the top of the roll-cage to rear mount of the vehicle (Figure 8).

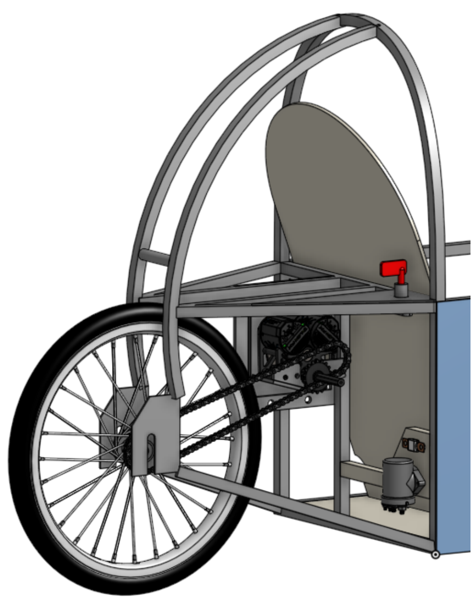

Figure 8: Back Section Design.

805 is equipped with 19-in diameter bicycle wheel, including the two forward wheel and rear wheel. This design provides significant amount of torque transfer from motor to the ground. Furthermore, the design of bicycle wheel reduces a significant amount of frictional force, allowing 805 to operate efficiently.

3.2 Steering System

3.3 Drivetrain

Overview

Due to the unsatifactory results in October Race, the team has come to a conclusion that 804 lacks the efficiency and energy in order to finish the race. Therefore, a motor that is designed for 12V race should be used on 805.

After researches, the design team has found a few suitable industrial motors that operates at 12V. Yet, none of the motor manufacturer replied the message, and such path was stalled.

Robotics motor has become the only option for the team. Consulting with serval FIRST robotics team members at the club, the Kraken X60s shows the best and is considered as the optimal motor for our situation. Other alternate motors include Kraken X44, NEO Vortex, NEO 2.0.

Performance Data

From data aquired from online resources, the following graph is created in order to determine the operation RPM, torque, and current.

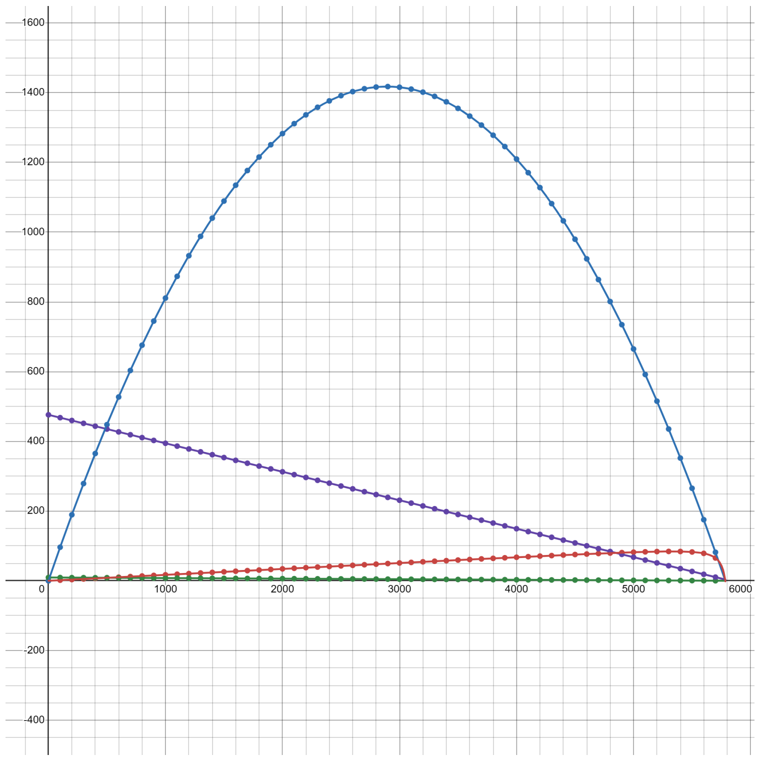

Figure 9: Kraken X60 Performance.

The x-axis is the RPM value; the blue line indicates the Power (W); the purple line indicates the current consumed (A); the green line indicates the Torque (Nm); the red line indicates the Efficiency.

The following link describes the above graph:
`https://www.desmos.com/calculator/baupqlsyxa`

Calculations: Maximum Current Delivers

A prelimary calculation is needed to determine the maximum currrent that can be delievred to the motor, in order to successfully finish the race.

The 12V Race requires 805 to equip with a single MTX-35 battery. MTX-35 is regarded as lead-acid battery. The following are specifications of MTX-35:

1. Weight: 18.1 kg
2. Reserve Capacity@25 Amps: 100 minutes
3. Voltage: 12V
4. Capacity: 55Ah
5. Maximum Amperage Intensity: 800 A
6. AGM Battery (Lead-Acid)

Instead of Watt's Law that governs the general battery performance, Lead-Acid battery are more accurately represented through the Peukert's Law, which is as follows:

$$C_p = I^k \times T \tag{1}$$

C_p is the capacity at a one-ampere discharge rate, which must be expressed in ampere hours; I is the actual discharge current (i.e.current drawn from a load) in amperes; t is the actual time to discharge the battery, which must be expressed in hours; k is the Peukert constant (dimensionless).

To find the Peukert's in the race situation, Equation 2 is used to determine the Peukert's constant (k).

$$k = \frac{\log(t_2) - log(t_1)}{log(I_1) - log(I_2)} \tag{2}$$

The two discharging values can be found through the 20-hour rate and RC rate. The 20-hour rate is the capacity listed (55Ah), where current is 2.75A by simple division. Reserve Capacity shows the dischanging rate under high load, which is 1.67 hr at 25A.

Subsituting these into the equation,

$$k = \frac{\log(1.67) - \log(20)}{\log(2.75) - \log(25)}$$
$$k = 1.125$$

With Peukert Constant for MTX-35, we can use equation 1 to determine the maximum current can be drawn continuously throughout the race. Assume the time is at 75 minutes (1.25), with 5 minutes of energy reserve.

$$C_p = I^k \times T$$

The capacity can be expressed through the 20-hour rate, since the Peukert Capacity is unknown,

$$2.75^{1.125} \times 20 = I_{race}^{1.125} \times 1.25$$
$$I_{race} = 33.2A$$

Therefore, 805 can only operate at a maximum current of 33.2 A continuously.

Motor Performance At Maximum Current

Since 805 is designed with two Kraken X60s connected in one flipped gearbox and has a parallel connection, the current of the motor is halved to 16.6 A continuous.

From Figure 9's desmos graph, one Kraken X60's performance at 16.6 A is as below:

1. Power: 158.8W
2. Efficiency: 76.7%
3. RPM: 5624 RPM
4. Torque: 0.258 Nm

As a parallel-connected motor system, the following is the final performance from the two motors:

1. Power: 317.6W
2. Efficiency: 76.7%
3. RPM: 5624 RPM
4. Torque: 0.516 NM

The gearbox purchased has a gear ratio of 9 : 1, which is further conenected a chain system of 24:18. The final gear ratio is 12:1.

According to the Inverse Law between Torque and Speed,

$$\frac{\tau_{input}}{\tau_{output}} = \frac{\text{Gear input}}{\text{Gear output}} \tag{3}$$
$$\frac{0.516}{\tau_{output}} = \frac{1}{12}$$
$$\tau_{output} = 6.192 \text{ Nm}$$

$$\frac{RPM_{input}}{RPM_{output}} = \frac{\text{Gear output}}{\text{Gear input}} \tag{4}$$
$$\frac{5624}{RPM_{output}} = \frac{12}{1}$$
$$RPM_{output} = 468.7$$

Therefore, the wheel has an output speed of 468.7 RPM, with 6.192 Nm.

3.4 Strategy

4 Results

5 Discussion

6 Conclusion